

# A two-term fermionic representation of the entire positivity core at $d = 8$ : Warnaar's finite forms land verbatim on Uncu's $S_{11}$ series

Seed 2, Layer 4, Round 2 (loop experiment 2026-07-04)

2026-07-04

## Abstract

We prove that for *every* profile  $c$  in the  $d = 8$  positivity core  $\{(6, 1, 1), (5, 2, 1), (5, 1, 2), (4, 3, 1), (4, 1, 3), (4, 2, 2)\}$  (and also for the already-settled balanced profile  $(3, 3, 2)$ ), the generating function  $G_c(z, q) = (zq; q)_\infty F_c(z, q)$  equals an explicit *difference of two manifestly nonnegative fermionic multisums*,  $G_c = \text{Ferm}_{c_2+1, c_3+1} - q \text{Ferm}_{c_2, c_3}$ , where the  $\text{Ferm}_{s,t}$  are the  $n_0, m_0 \rightarrow \infty$  limits of Warnaar's proved finite-form family  $F_{n_0, m_0; 3, s, t}^{(-1)}$ . The key observation is that these limits coincide *verbatim* with Uncu's series  $S_{11}(z; e_{s-1} | e_{t-1})$  — including the  $(q; q)_{r_3+s_3+1}$  denominator — so no Kanade–Russell contiguous relation and no Pochhammer merge is needed. Consequently  $Q_{n,c}(q) = (q; q)_n [z^n] G_c$  is, for each core profile, an explicit finite  $q$ -binomial two-term difference, and Warnaar's positivity conjecture at the entire  $d = 8$  core is reduced to six explicit polynomial inequalities. This covers in particular  $(4, 2, 2)$ , which appears in no  $R$ -relation at  $d = 8$  and was previously unreachable by any known route.

## 1 Setup

Notation as in the Layer-3 write-up (prove-seed2-layer3.tex):  $F_c(z, q)$  is the Corteel–Welsh cylindric-partition generating function of profile  $c = (c_1, c_2, c_3)$ ,  $c_1 + c_2 + c_3 = d = 8$  (modulus  $m = 3k - 1 = 11$ ,  $k = 4$ ),

$$G_c = (zq; q)_\infty F_c, \quad H_c = G_c / (q; q)_\infty, \quad Q_{n,c} = (q; q)_n [z^n] G_c,$$

and for  $\rho, \sigma \in \mathbb{Z}^3$ ,

$$S_{11}(z; \rho | \sigma) = \sum_{\substack{r_1 \geq r_2 \geq r_3 \geq 0 \\ s_1 \geq s_2 \geq s_3 \geq 0}} z^{r_1} \frac{q^{\sum_{i=1}^3 (r_i^2 - r_i s_i + s_i^2 + \rho_i r_i + \sigma_i s_i) + 2r_3 s_3}}{(q; q)_{r_1 - r_2} (q; q)_{r_2 - r_3} (q; q)_{r_3} (q; q)_{s_1 - s_2} (q; q)_{s_2 - s_3} (q; q)_{s_3} (q; q)_{r_3 + s_3 + 1}}, \quad (1)$$

with  $e_3 = (0, 0, 0)$ ,  $e_2 = (0, 0, 1)$ ,  $e_1 = (0, 1, 1)$ ,  $e_0 = (1, 1, 1)$ . Uncu's modulo-11 list (2024, eq. mod11list; each row proved by his Theorem m11, itself computer-assisted via the  $N = 6$  finite verification of thm:n6m11) states, for each of the seven listed profiles  $(3, 3, 2)$ ,  $(6, 1, 1)$ ,  $(5, 2, 1)$ ,  $(5, 1, 2)$ ,  $(4, 3, 1)$ ,  $(4, 1, 3)$ ,  $(4, 2, 2)$ ,

$$H_{(c_1, c_2, c_3)}(z, q) = S_{11}(e_{c_2} | e_{c_3}) - q S_{11}(e_{c_2-1} | e_{c_3-1}). \quad (2)$$

Warnaar (A<sub>2</sub> Andrews–Gordon paper, eq. (F2) of Section 7) defines, for  $1 \leq s, t \leq k+1$  and  $\sigma = (1, \dots, 1, a)$ ,

$$F_{n_0, m_0; k, s, t}^{(a)}(z, q) := \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_k \geq 0}} \frac{z^{n_1} q^{\sum_{i=s}^k n_i + \sum_{i=t}^k m_i}}{(q; q)_{n_k + m_k + 1}} \prod_{i=1}^k q^{n_i^2 - \sigma_i n_i m_i + m_i^2} \begin{bmatrix} n_{i-1} \\ n_i \end{bmatrix} \begin{bmatrix} m_{i-1} \\ m_i \end{bmatrix}, \quad (3)$$

and *proves* (his Proposition on the second finite form, via his Lemma on the  $\mathcal{F}$ -transformation; this is unconditional — the separate conjecture on missing ASW theta-evaluations is not used here) the polynomial identities: for  $a = -1$ ,  $1 \leq s \leq k+1$ ,  $1 \leq t \leq k$ , with  $m_k := 2n_k$ ,

$$F_{n_0, m_0; k, s, t}^{(-1)}(z, q) = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{n_k^2 + \sum_{i=s}^k n_i + \sum_{i=t}^{k-1} m_i}}{(q; q)_{m_0 - n_1 + m_1 + \delta_{t,1}}} \begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \\ \times \prod_{i=1}^{k-1} q^{n_i^2 - n_i m_i + m_i^2} \begin{bmatrix} n_i \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} + \delta_{i,t-1} \\ m_i \end{bmatrix}, \quad (4)$$

and for  $a = -1$ ,  $1 \leq s \leq k$ ,  $t = k+1$ ,  $k \geq 2$ ,

$$F_{n_0, m_0; k, s, k+1}^{(-1)}(z, q) = \sum_{\substack{n_1, \dots, n_k \geq 0 \\ m_1, \dots, m_{k-1} \geq 0}} \frac{z^{n_1} q^{n_k^2 - n_k + \sum_{i=s}^k n_i}}{(q; q)_{m_0 - n_1 + m_1}} \begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \\ \times \prod_{i=1}^{k-1} q^{n_i^2 - n_i m_i + m_i^2} \begin{bmatrix} n_i + \delta_{i,k-1} \\ n_{i+1} \end{bmatrix} \begin{bmatrix} n_i - n_{i+1} + m_{i+1} \\ m_i \end{bmatrix}. \quad (5)$$

## 2 The verbatim limit

**Lemma 1** (Verbatim match). *Fix  $k = 3$ ,  $a = -1$ ,  $1 \leq s \leq 4$ ,  $1 \leq t \leq 4$ ,  $(s, t) \neq (4, 4)$ . Then, coefficientwise in  $z$  and  $q$ ,*

$$\lim_{n_0, m_0 \rightarrow \infty} F_{n_0, m_0; 3, s, t}^{(-1)}(z, q) = S_{11}(z; e_{s-1} | e_{t-1}),$$

where  $e_{s-1}, e_{t-1}$  are as above (i.e.  $(e_j)_i = 1$  iff  $i > j$ ).

*Proof.* In (3) the binomial chains force  $n_1 \geq n_2 \geq n_3$  and  $m_1 \geq m_2 \geq m_3$ , and as  $n_0, m_0 \rightarrow \infty$ ,

$$\begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \longrightarrow \frac{1}{(q; q)_{n_1 - n_2} (q; q)_{n_2 - n_3} (q; q)_{n_3}},$$

coefficientwise (each coefficient stabilizes), and likewise for the  $m$ -chain. The quadratic exponent with  $\sigma = (1, 1, -1)$  is  $\sum_i (n_i^2 - n_i m_i + m_i^2) + 2n_3 m_3$ , the linear exponent is  $\sum_{i \geq s} n_i + \sum_{i \geq t} m_i = e_{s-1} \cdot n + e_{t-1} \cdot m$ , and the denominator  $(q; q)_{n_3 + m_3 + 1}$  is precisely the  $(q; q)_{r_3 + s_3 + 1}$  of (1). Renaming  $(n, m) \mapsto (r, s)$  gives the claim.  $\square$

**Lemma 2** (Fermionic form of a single  $S_{11}$  term). *For  $1 \leq s \leq 4$ ,  $1 \leq t \leq 3$ , define the manifestly nonnegative sum*

$$\text{Ferm}_{s,t}(z, q) := \sum_{\substack{n_1 \geq n_2 \geq n_3 \geq 0 \\ m_1, m_2 \geq 0}} \frac{z^{n_1}}{(q; q)_{n_1}} q^{n_3^2 + \sum_{i=s}^3 n_i + \sum_{i=t}^2 m_i + \sum_{i=1}^2 (n_i^2 - n_i m_i + m_i^2)} \quad (6)$$

$$\times \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \begin{bmatrix} n_1 - n_2 + m_2 + \delta_{t,2} \\ m_1 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \begin{bmatrix} n_2 + n_3 + \delta_{t,3} \\ m_2 \end{bmatrix},$$

and for  $t = 4$  (with  $1 \leq s \leq 3$ )

$$\text{Ferm}_{s,4}(z, q) := \sum_{\substack{n_1 \geq n_2 \geq 0, 0 \leq n_3 \leq n_2 + 1 \\ m_1, m_2 \geq 0}} \frac{z^{n_1}}{(q; q)_{n_1}} q^{n_3^2 - n_3 + \sum_{i=s}^3 n_i + \sum_{i=1}^2 (n_i^2 - n_i m_i + m_i^2)} \quad (7)$$

$$\times \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \begin{bmatrix} n_1 - n_2 + m_2 \\ m_1 \end{bmatrix} \begin{bmatrix} n_2 + 1 \\ n_3 \end{bmatrix} \begin{bmatrix} n_2 + n_3 \\ m_2 \end{bmatrix}.$$

Then  $S_{11}(z; e_{s-1} | e_{t-1}) = \text{Ferm}_{s,t}(z, q) / (q; q)_\infty$ .

*Proof.* Take  $n_0, m_0 \rightarrow \infty$  in (4) (resp. (5)) at  $k = 3$ :  $\begin{bmatrix} n_0 \\ n_1 \end{bmatrix} \rightarrow 1/(q; q)_{n_1}$  and  $1/(q; q)_{m_0 - n_1 + m_1 + \delta_{t,1}} \rightarrow 1/(q; q)_\infty$  coefficientwise (the sums over  $m_2, m_1$  are finite for each fixed  $(n_1, n_2, n_3)$ , so the limit may be taken termwise), with  $m_3 := 2n_3$  substituted in the  $i = 2$  binomial,  $\begin{bmatrix} n_2 - n_3 + m_3 + \delta_{2,t-1} \\ m_2 \end{bmatrix} = \begin{bmatrix} n_2 + n_3 + \delta_{t,3} \\ m_2 \end{bmatrix}$ . Combine with Lemma 1.  $\square$

### 3 Main theorem

**Theorem 3** (Two-term fermionic representation of the  $d = 8$  core). *For each of the seven profiles  $(6, 1, 1)$ ,  $(5, 2, 1)$ ,  $(5, 1, 2)$ ,  $(4, 3, 1)$ ,  $(4, 1, 3)$ ,  $(4, 2, 2)$  and  $(3, 3, 2)$  — i.e. cyclic representatives of all six remaining core orbits together with the settled balanced orbit —*

$$G_c(z, q) = \text{Ferm}_{c_2+1, c_3+1}(z, q) - q \text{Ferm}_{c_2, c_3}(z, q).$$

Consequently, for all  $n \geq 0$ ,

$$Q_{n,c}(q) = \text{ferm}_{c_2+1, c_3+1}(n) - q \text{ferm}_{c_2, c_3}(n),$$

where  $\text{ferm}_{s,t}(n) := (q; q)_n [z^n] \text{Ferm}_{s,t}$  is the finite nonnegative  $q$ -binomial multisum obtained from (6)–(7) by setting  $n_1 = n$  and cancelling  $(q; q)_n$  against the denominator.

*Proof.* By Lemmas 1 and 2,  $S_{11}(e_{c_2} | e_{c_3}) = \text{Ferm}_{c_2+1, c_3+1} / (q; q)_\infty$  and  $S_{11}(e_{c_2-1} | e_{c_3-1}) = \text{Ferm}_{c_2, c_3} / (q; q)_\infty$  (the parameter ranges are admissible for each of the seven listed profiles: all thirteen  $(s, t)$  instances that occur satisfy  $1 \leq s \leq 4$ ,  $1 \leq t \leq 4$ ,  $(s, t) \neq (4, 4)$ ; the only  $t = 4$  instance is  $S(e_1 | e_3)$  for  $c = (4, 1, 3)$ , with  $s = 2 \leq 3$ . Note the restriction to the listed profiles is necessary: e.g.  $(2, 3, 3)$  would require  $(s, t) = (4, 4)$ , covered by neither (4) nor (5), and  $(1, 6, 1)$  would require the undefined  $\text{Ferm}_{7,2}$ ). Multiply Uncu's proved identity (2) by  $(q; q)_\infty$  and use  $G_c = (q; q)_\infty H_c$ . The  $Q$ -level statement follows because the  $z^n$ -coefficient of  $\text{Ferm}_{s,t}$  carries the single denominator  $(q; q)_n$ .  $\square$

**Corollary 4.** *Warnaar’s positivity conjecture at the entire  $d = 8$  core is equivalent to the six families of explicit polynomial inequalities*

$$\text{ferm}_{c_2+1, c_3+1}(n) \geq q \text{ferm}_{c_2, c_3}(n) \quad (n \geq 0)$$

for the six core orbits. In particular this includes  $(4, 2, 2)$ , which appears in no Kanade–Russell  $R$ -relation at  $d = 8$ .

**Remark 5** (Numerical verification). *All links were verified in exact arithmetic (`seed2_R2L4.verify.sage`):*

(i) the transcription of (3)–(5) as an exact  $z$ -graded power-series identity at  $k = 3$ ,  $a = -1$ , for all fifteen admissible  $(s, t)$  and all  $n_0, m_0 \in \{0, \dots, 3\}$  (to  $q^{150}$ ); (ii) Lemmas 1/2 at  $z$ -orders  $n \leq 3$  for six spot pairs  $(s, t)$  including a  $t = 4$  case; (iii) Theorem 3 at the  $Q$ -level: the two-term difference equals the raw-validated reference engine’s  $Q_{n,c}$  (target-first kernel, Gauss inversion) exactly in  $\mathbb{Z}[q]$  for all seven profiles and all  $n \leq 12$ , and is coefficientwise nonnegative throughout that range.

**Remark 6** (Structure of the remaining inequalities). Write  $Q_F = n^2 + n_2^2 + n_3^2 - nm_1 - n_2m_2 + m_1^2 + m_2^2$ ,  $N = n - n_2 + m_2$ ,  $M = n_2 + n_3$ . For the three orbits with  $c_3 = 1$  the difference collapses to a single-row shape

$$Q_{n,c} = \sum_{n_2, n_3, m_2} q^{Q_F+w} \begin{bmatrix} n \\ n_2 \end{bmatrix} \begin{bmatrix} n_2 \\ n_3 \end{bmatrix} \begin{bmatrix} M \\ m_2 \end{bmatrix} \sum_{m_1} q^0 \left( \begin{bmatrix} N+1 \\ m_1 \end{bmatrix} - q^{1+n_{c_2}+m_1} \begin{bmatrix} N \\ m_1 \end{bmatrix} \right),$$

with penalty exponent  $1 + n_{c_2} + m_1$  (where  $n_1 := n$ ) and  $w = n_2 + n_3 + m_2$ ,  $n_3 + m_2$ ,  $m_2$  for  $c = (6, 1, 1), (5, 2, 1), (4, 3, 1)$  respectively. Since  $\begin{bmatrix} N+1 \\ m_1 \end{bmatrix} = q^{m_1} \begin{bmatrix} N \\ m_1 \end{bmatrix} + \begin{bmatrix} N \\ m_1-1 \end{bmatrix}$ , the  $(6, 1, 1)$  case (where  $n_{c_2} = n$  is constant in the sum) takes the exact  $d = 4$  wall shape  $Q_n = (1 - q^{n+1})A_n + B_n$  with  $A_n, B_n$  manifestly nonnegative. Slice experiments (exact,  $n \leq 7$ ) show: for  $(4, 3, 1)$  every fixed- $(n_2, n_3)$  partial sum is already nonnegative; for  $(5, 2, 1)$  every fixed- $n_2$  partial sum is nonnegative; for  $(6, 1, 1)$  only the full sum is. A one-round absorption residual  $B - q^{1+n_3}A$  is not nonnegative for  $(4, 3, 1)$ , so the required absorption lemma is genuinely multi-row; proving these six inequalities is the remaining open step for  $d = 8$ .

**Remark 7** (Relation to the Layer-3/Seed-1 chains). The  $(3, 3, 2)$  theorem (Layer 3) used the first finite-form family (Ferm without linear terms), one Pochhammer split and one Kanade–Russell relation  $R_3$ ; the  $d = 10$  balanced theorem (Seed 1, Layer 4) found that at modulus 13 the split lands verbatim on Uncu’s pair. The present result completes the pattern in a stronger form: at modulus 11 Warnaar’s second finite-form family  $F_{3,s,t}^{(-1)}$  hits every single  $S_{11}(e_a|e_b)$  term verbatim (the  $+1$  in its  $(q; q)_{n_k+m_k+1}$  denominator is exactly Uncu’s  $(q; q)_{r_3+s_3+1}$ ), so the entire modulo-11 list of Uncu  $S$ -pairs — hence the entire  $d = 8$  core — acquires proved two-term fermionic representations with no bridging relations at all. The depth- $\leq 4$   $R_1/R_2$  word search that was Mission 2’s stated mechanism is thereby superseded for representation purposes.